



GOLDLEVEL SCIENCE RESEARCH ARTIFACT

Colibactin and the Early-Onset Colorectal Cancer Question

A public evidence read on mutational signatures, microbiome exposure, and causal boundaries

Public read

Colorectal cancer is rising among younger adults. A 2025 Nature study adds an important genomic clue: some early-onset colorectal cancers carry DNA damage patterns linked to colibactin, a mutagen produced by certain gut bacteria.

This does not solve the cause. It sharpens the question.

Key idea

The study is not a declaration of causation. It is a map of mutational evidence that may point toward early-life exposure.

Version 1.0.0 | Public Evidence Read | 2026

Why this study matters

Colorectal cancer is not only a disease of older age. Across many countries, more adults under 50 are being diagnosed, and the reasons remain unclear.

That uncertainty matters. If a disease pattern changes across generations, geography, and age groups, the public question becomes larger than individual lifestyle alone. Researchers need ways to detect whether hidden exposures have left measurable biological traces.

Definition

Early-onset colorectal cancer usually refers to colorectal cancer diagnosed before age 50.

Evidence note

The Nature paper reports that early-onset colorectal cancer has doubled in many countries over the past two decades and that the reasons for the rise are unknown. [1]

Key idea

The study does not start with a known cause. It starts with a pattern written into cancer genomes.

Caution

Rising incidence does not automatically identify a cause. It defines the public health question.

What the study actually did

The researchers examined colorectal cancer genomes from 11 countries across four continents. Instead of asking only where cancer is more common, they asked whether the DNA of tumours shows different patterns of mutation by geography and age.

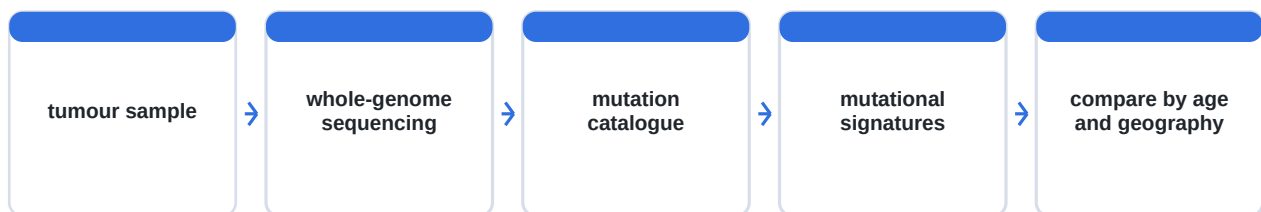
This is called a mutational-signature approach: looking for repeated DNA damage patterns that may point back to the process that created them.

Definition

A mutational signature is a repeated pattern of DNA changes. Some signatures are linked to known processes, while others remain unexplained.

Evidence note

The study examined 981 colorectal cancers, including 132 early-onset cases. Its main analysis focused on 802 microsatellite-stable / DNA repair-proficient colorectal cancers. [1]



Key idea

The method turns cancer genomes into a record of past biological damage.

The genomic fingerprint: SBS88 and ID18

The study's most public-facing signal concerns two mutational signatures: SBS88 and ID18. These are linked to exposure to colibactin, a DNA-damaging compound produced by certain bacteria that can carry the pks island.

The researchers found that these signatures were more common in early-onset colorectal cancers than in later-onset cases.

Definition

Colibactin is a bacteria-produced mutagen: a compound capable of damaging DNA.

Evidence note

The paper reports that SBS88 and ID18 were enriched in early-onset colorectal cancers and were 3.3 times more common in cancers diagnosed before age 40 than in those diagnosed after age 70. [1]

Key idea

The finding is a DNA trace, not a simple one-cause answer.

Caution

A mutational signature shows that a mutational process occurred. It does not, by itself, prove that this process explains the whole population-level rise.

Why geography matters

The study compared cancers from multiple countries because cancer risk is not evenly distributed across the world. Different countries can carry different environmental, lifestyle, microbial, dietary, healthcare, and inherited-risk patterns.

The paper found that several mutational signatures varied by country. Some had unknown causes. This makes the study larger than colibactin alone: it shows how cancer genomes can reveal geographically patterned mutational processes.

Context note

A geographic signal does not automatically mean one exposure is responsible. It means the pattern is uneven enough to investigate.

**Key idea**

Geography turns the cancer genome into an exposure map - but the map still needs interpretation.

**GOLDLEVEL
read**

The safest public framing is: the study opens multiple exposure questions, with colibactin as one of the strongest age-linked signals.

The cause model: plausible, not proven

The strongest evidence-bounded model is not "colibactin explains everything." The stronger model is more precise:



Principle

A cause model is not the same as proof. It is a structured hypothesis that must survive further testing.

Key idea

The study supports a plausible contributor model, not a completed causal verdict.

Warning

Do not collapse this into "gut bacteria cause young colon cancer." That is too broad and not what the paper proves.

Early imprint, later disease

One reason this study matters is timing. The paper reports that SBS88 and ID18 were enriched in early clonal mutations, meaning these signatures appeared early in the development of the tumour.

That matters because a mutation does not need to happen near diagnosis to affect risk. A DNA damage event earlier in life may alter the future trajectory of a cell lineage.

Definition

An early clonal mutation is a mutation present early enough that later tumour cells inherit it.

Evidence note

SBS88 and ID18 were enriched in early clonal compared with late clonal mutations, supporting the interpretation that the imprint occurred early in tumour evolution. [1]

Key idea

A biological event can matter decades later if it changes the starting conditions of tissue evolution.

Plain-language translation

The question is not only "What happened before diagnosis?" It may also be "What happened early enough to give some cells a head start?"

Why current bacteria may not show the whole history

The study did not find a simple match between current detection of pks+ bacteria and the presence of SBS88 or ID18 signatures in the tumour genome.

This is important. It means the mutational imprint may reflect an earlier exposure that is no longer visible through current microbial detection. The microbiome changes over time; the DNA imprint can remain.

Current microbiome test	a snapshot
Tumour mutational signature	a historical trace

Evidence note

The paper reports no association between the presence of SBS88/ID18 and current pks+ bacteria detection, and suggests one explanation is earlier-life imprinting followed by microbiome change over time. [1]

Key idea

Absence of current bacterial detection does not erase a past mutational imprint.

Caution

This also means the study does not create a ready-made public screening test for colibactin exposure.

How colibactin could matter biologically

For a mutational process to matter in cancer development, it must be able to affect important genes or pathways. The study found a link between colibactin-associated signatures and driver mutations in APC, a major colorectal cancer driver gene.

In colibactin-positive cancers, ID18 was responsible for about one quarter of APC driver indels.

Definition

A driver mutation is a mutation that helps a cancer grow or progress, rather than a random passenger change.

Evidence note

The paper reports that ID18 accounted for about 25.3 percent of APC driver indels in colibactin-positive cancers. [1]

Key idea

The APC link gives the colibactin hypothesis biological plausibility.

Plain-language translation

If colibactin damages DNA in the wrong place, early enough, it may help start one pathway toward cancer in some cases.

Warning

Biological plausibility is not the same as population-level proof.

The claim boundary

This study is important because it strengthens a prevention question. It does not settle the causal answer.

The authors explicitly state that further research is required to establish causality. Future work needs normal colorectal tissue, age-matched controls, stronger microbiome analysis, and prevention evidence.

Evidence note

The paper states that further research is required to establish causality, and that definitive evidence would require reduced cancer incidence after prevention of early-life exposure to colibactin-producing bacteria. [1]

Do not claim:

- Colibactin is the cause of young-onset colorectal cancer.
- E. coli causes colorectal cancer.
- The microbiome alone explains the trend.
- A clinical microbiome test should now be used by the public.
- A proven prevention strategy already exists.
- Diet, BMI, heredity, lifestyle, healthcare access, and other exposures no longer matter.

Key idea

Potential contributor is evidence-safe. Cause found is not.

Warning

The public value of this artifact depends on preserving the uncertainty. Overclaiming would damage the signal.

Why the public should understand this

This paper sits at the crossing point of cancer epidemiology, microbiome science, geography, genomics, early-life exposure, and prevention research.

Its public value is not that it gives people a new fear. Its value is that it shows how cancer genomes can reveal hidden exposure histories - and how prevention science may need to look earlier in life than diagnosis.

Why this matters

If early-life microbial exposures can leave long-term mutational consequences, prevention research may eventually need to include microbiome ecology, childhood exposures, and population-level environmental change.



Key idea

The study does not give the public a panic message. It gives science a sharper prevention question.

GOLDLEVEL read: the cause is not solved, but the signal is real

The strongest interpretation is not:

Not the claim Colibactin explains everything.

The strongest interpretation is:

GOLDLEVEL read Some early-onset colorectal cancers may carry an early-life bacterial mutagenic imprint.

That is a narrower claim. It is also more useful. It protects the science from exaggeration while preserving the importance of the signal.

Young-onset colorectal cancer is rising	Supported
SBS88/ID18 are enriched in younger cases	Supported
SBS88/ID18 appear early in tumour development	Supported
Colibactin may contribute to some early-onset cases	Plausible hypothesis
Colibactin causes the population-level rise	Unproven
A prevention strategy exists now	Unproven

Key idea The role of a public evidence read is not to amplify certainty. It is to stabilise attention around what the evidence can actually bear.

The public takeaway

A 2025 Nature study found that colibactin-linked mutation patterns were enriched in early-onset colorectal cancers and appeared early in tumour development.

This does not prove causality. It does support a serious prevention question: whether exposure to colibactin-producing bacteria early in life may leave mutational imprints that contribute to earlier cancer development in some people.

In summary

- The rise in young-onset colorectal cancer remains unexplained.
- The study found age- and geography-linked mutational patterns.
- SBS88 and ID18 are the key colibactin-linked signatures.
- Their timing suggests an early imprint.
- Their APC link gives biological plausibility.
- Causality remains unproven.
- The public message should be curiosity, not alarm.

Key idea

The signal is not a final answer. It is a better question.

Source status and public-use limits

This artifact separates observed findings, interpretation, and unproven causality. That separation is part of the public-facing method.

Known / observed

- The primary source is the 2025 Nature paper by Diaz-Gay et al.
- The study reports 981 colorectal cancer genomes from 11 countries.
- The main mutational-signature analysis focused on 802 MSS / DNA repair-proficient cancers.
- The study reports SBS88 and ID18 enrichment in early-onset colorectal cancers.
- The study reports early clonal enrichment for SBS88 and ID18.
- The study reports ID18 contribution to APC driver indels in colibactin-positive cases.
- The authors state that further research is required to establish causality.

Inferred

- The evidence supports a contributor hypothesis rather than a sole-cause model.
- A public artifact is justified because the findings are important but easy to overstate.
- GOLDLEVEL is positioned here as an evidence-reading layer, not a medical authority.

Unproven / blocked

- Colibactin as the proven cause of the population-level rise.
- Any immediate public microbiome testing recommendation.
- Any individual diagnosis, screening, treatment, or prevention instruction.

Warning

This artifact is public science interpretation only. It is not medical advice and must not be used for diagnosis, treatment, screening, or individual health decisions.

References

Primary reference

1. Diaz-Gay, M. et al. Geographic and age variations in mutational processes in colorectal cancer. *Nature* 643, 230-240 (2025). <https://doi.org/10.1038/s41586-025-09025-8>

2. Bray, F. et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J. Clin.* 74, 229-263 (2024).

3. Siegel, R. L., Jemal, A. & Ward, E. M. Increase in incidence of colorectal cancer among young men and women in the United States. *Cancer Epidemiol. Biomarkers Prev.* 18, 1695-1698 (2009).

4. Vuik, F. E. et al. Increasing incidence of colorectal cancer in young adults in Europe over the last 25 years. *Gut* 68, 1820-1826 (2019).

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6. Patel, S. G. et al. The rising tide of early-onset colorectal cancer: a comprehensive review of epidemiology, clinical features, biology, risk factors, prevention, and early detection. *Lancet Gastroenterol. Hepatol.* 7, 262-274 (2022).

7. Lee-Six, H. et al. The landscape of somatic mutation in normal colorectal epithelial cells. *Nature* 574, 532-537 (2019).

8. Pleguezuelos-Manzano, C. et al. Mutational signature in colorectal cancer caused by genotoxic pks+ *E. coli*. *Nature* 580, 269-273 (2020).

Reference status	References 2-8 are supporting references cited within the primary Nature paper. They are included to preserve the source chain behind the public evidence read.
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Version and claim-control record

Field	Value
Artifact name	GOLDLEVEL Science Research Artifact - Colibactin CRC
Version	1.0.0
Status	Public-facing draft PDF
Primary evidence	Uploaded 2025 Nature article by Diaz-Gay et al.
Claim boundary	Association and plausible contributor model; causality not established.
Public role	Field harmoniser and stabiliser.

Warning

This PDF does not prove medical causality, clinical readiness, source-rights clearance beyond cited/public references, deployment, or real-world prevention effect.

Key idea

The cause is not solved, but the signal is real.

End of artifact.